

When Simple Wins: CNN vs. Transformer Encoders for Low-Data Nucleus Segmentation

CNN vs. Transformer in the Low-Data Regime

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Problem Setting

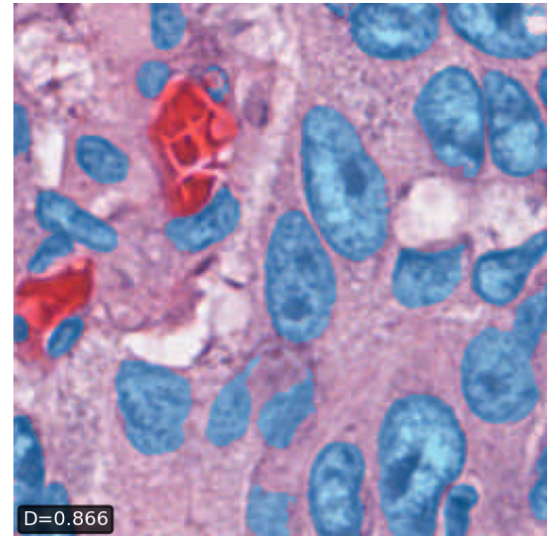
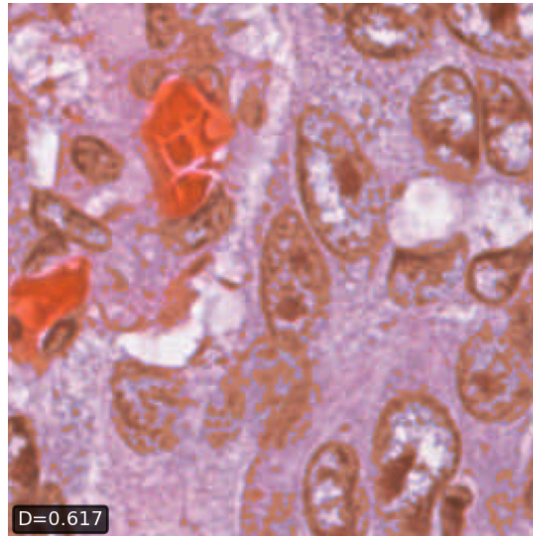
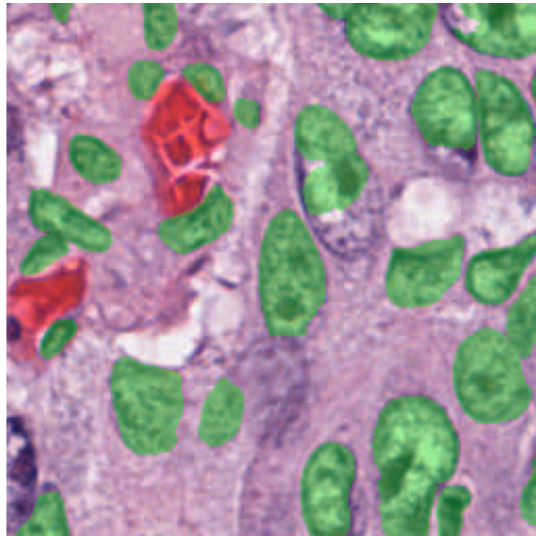
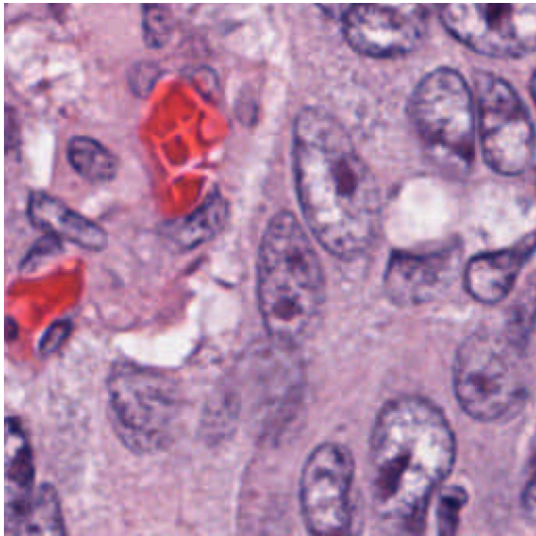
The gap: Pathology foundation models (UNI 307M, Virchow2 632M params) require large amounts of GPU memory for full fine-tuning, restricting most users to linear probing. This rules out edge-deployed microscopy scanners in rural clinics, mobile pathology units run by organizations like MSF on laptop hardware, under-resourced labs studying rare diseases or animal models on a single consumer GPU, and any non-H&E staining protocol where H&E-pretrained weights don't transfer regardless of scale.

Question: Among *lightweight, from-scratch* architectures, which encoder works best with limited data?

Input	256×256 H&E-stained histopathology patch
Output	Binary mask — nucleus vs. background
PanNuke	~5K training patches, 19 tissue types
MoNuSeg	37 training whole-slide images — genuinely small-data

Problem Setting — The Task

Input → Ground Truth → Classical baseline (Otsu) → Our best model (VGG)



Baseline & What We Built

Classical baseline — Otsu Thresholding (0 params) — Dice **0.482**

Our contribution: U-Net with a fixed decoder and shared encoder interface

The decoder, loss, optimizer, and training schedule are held constant.

The **encoder backbone is the only variable** — swapped via config.

Encoder	Architecture	Params	Pretrained?
VGG	Stacked 3×3 convs, max-pool	34M	No
ResNet	Residual blocks, stride-2 down	37.5M	No
Swin-T (scratch)	Shifted-window attention	86.8M	No
Swin-T (pretrained)	Shifted-window attention	86.8M	ImageNet

MedT (Valanarasu et al., MICCAI 2021) — gated axial-attention transformer built for small medical datasets — serves as the published baseline on MoNuSeg. Dice **0.796**, 1.4M params.

Results

PanNuke (fold 3)

Encoder	Dice ↑	IoU ↑	Params
Otsu (classical)	0.482	0.349	0
Swin-T (scratch)	0.590	0.442	86.8M
Swin-T (pretrained)	0.802	0.676	86.8M
ResNet	0.849	0.740	37.5M
VGG	0.851	0.744	34.0M

MoNuSeg test set

Model	Dice ↑	IoU ↑	Params
MedT (30 train imgs)	0.796	0.662	1.4M
Swin-T pretrained (37 imgs)	0.750	0.604	86.8M
VGG (37 imgs)	0.796	0.663	34.0M

Key Findings

(1) Transformers without pretraining barely learn useful features at this scale

Swin from scratch barely beats Otsu: **0.590 vs 0.482** Dice — without pretraining, transformers barely learn useful features at this data scale.

(2) Residual connections add nothing at 4 stages

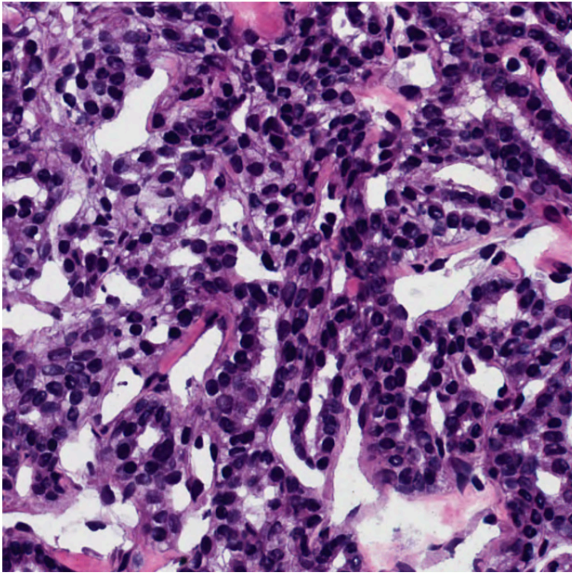
VGG = ResNet: **0.851 vs 0.849** — skip connections solve gradient degradation in *deep* networks; at 4 encoder stages neither architecture degrades.

(3) CNN matches purpose-built small-data transformer

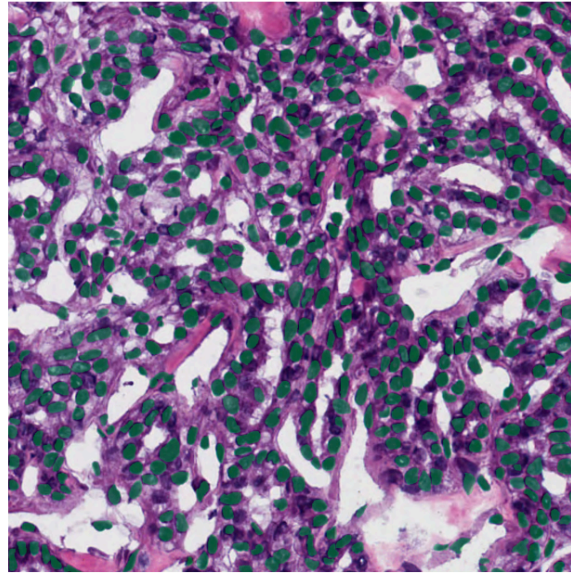
VGG matches MedT (**0.796 vs 0.796**) — a transformer with gated axial-attention designed specifically for small medical datasets — while outperforming pretrained Swin (0.750). Convolutional inductive bias is the decisive advantage in the low-data regime.

Results — MoNuSeg Qualitative (Input | Ground Truth | VGG | Swin)

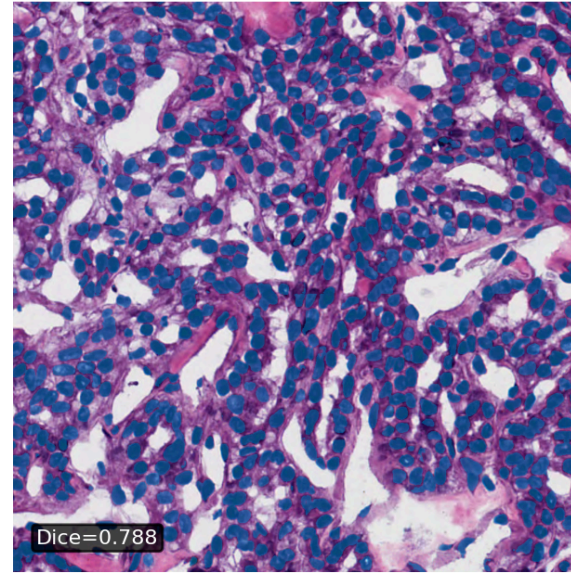
Input



Ground Truth



VGG



Swin (pretrained)

